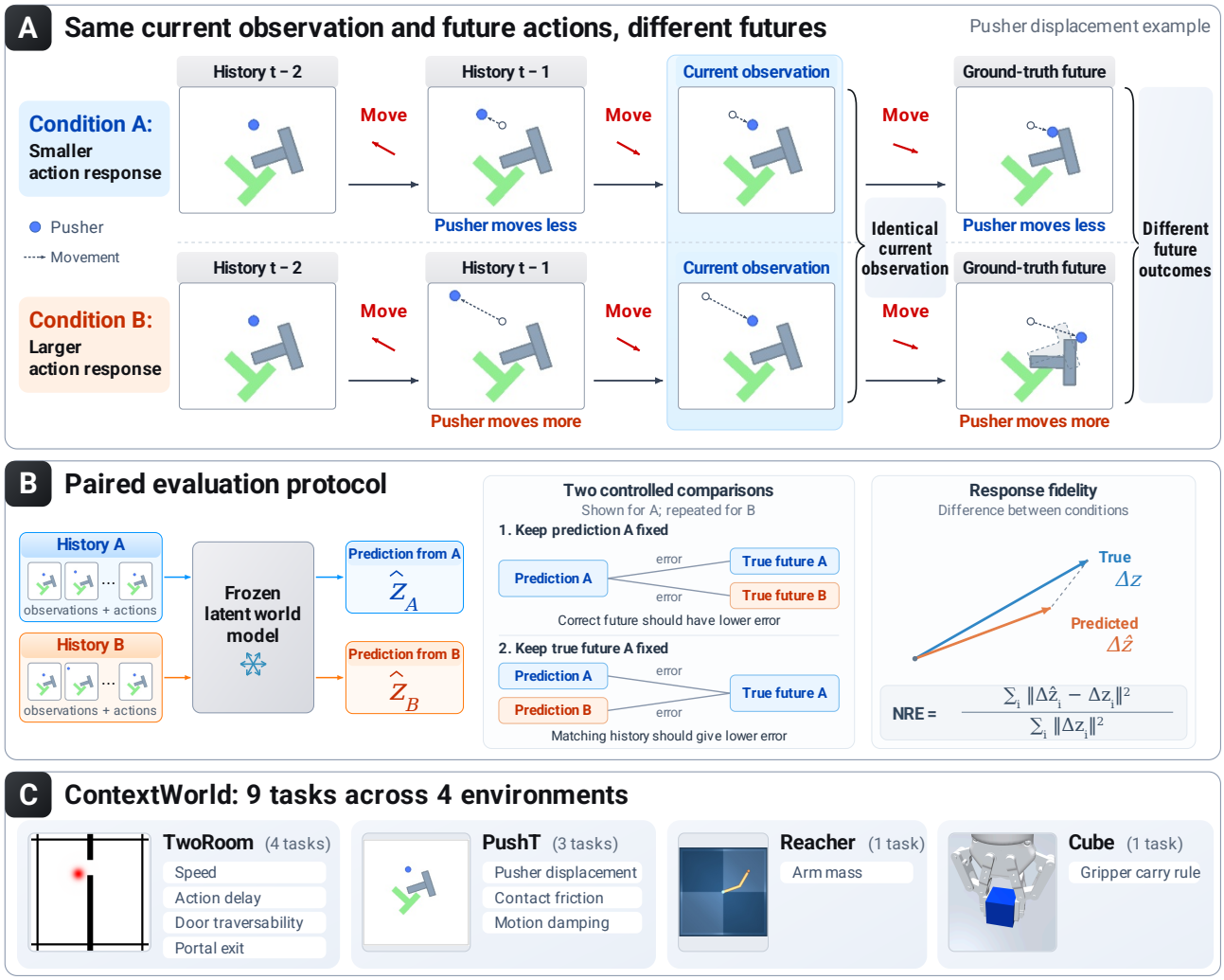
Overview with response fidelity



ContextWorld overview. (A) A recorded pusher-displacement pair. Shared actions reveal different movement distances in the history, return both conditions to the same current observation, and produce different future states. Red arrows summarize action directions; dashed arrows show observed endpoint displacement, including the return movement. The translucent outline in the lower future panel marks the T block's current pose. These overlays are annotations, not model inputs. All images use the same crop of the original observations. (B) A frozen model predicts a future from each history and the shared future actions. The two comparisons hold different quantities fixed. Future discrimination compares one prediction with the two true futures. History benefit compares the predictions from histories A and B with the same true future. The diagrams show condition A; both comparisons are repeated for B. Connecting lines denote latent prediction-error comparisons, and the text below each diagram states the desired outcome. Response fidelity compares the predicted difference $\Delta\hat{z} = \hat{z}^B - \hat{z}^A$ with the true difference $\Delta z = z^B - z^A$. NRE measures their discrepancy, with zero indicating an exact response. The arrows illustrate this comparison schematically; they are not a projection of measured model outputs. (C) Nine tasks across four environments.